

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Tool	1	hr

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9702/36

October/November 2014

2 hours

Additional Materials: As listed in the Confidential Instructions.

DO NOT WRITE IN ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

You may lose marks if you do not show your working or if you do not use appropriate units.

Additional answer paper and graph paper should be used only if it becomes necessary to do so.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	



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You may not need to use all of the materials provided.

1 In this experiment, you will investigate a network of resistors.

(a) (i) Assemble the circuit of Fig. 1.1.

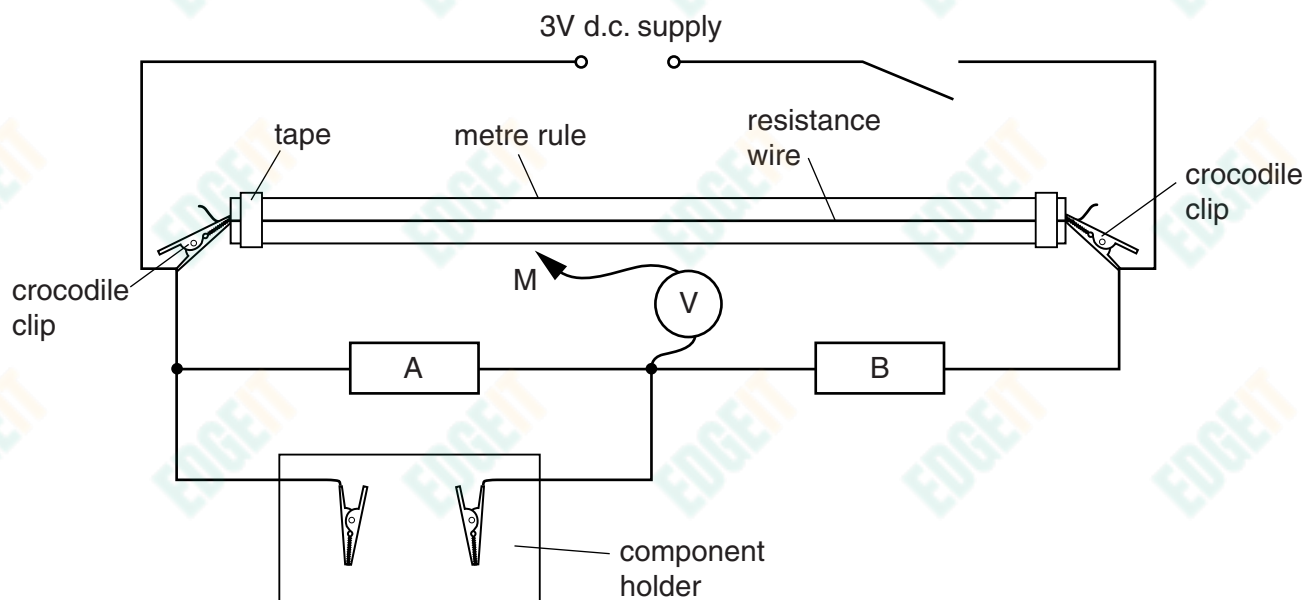


Fig. 1.1

(ii) To check that your circuit is correct, close the switch and place the movable contact M on the resistance wire at the 50cm mark. Record the voltmeter reading, which should be between 0.3V and 0.7V.

voltmeter reading = [1]

(iii) Open the switch.

- (b) (i) Select one of the resistors labelled with a numerical value and connect it between the crocodile clips of the component holder. Record the resistance R of your selected resistor.

$R = \dots\dots\dots \text{ k}\Omega$

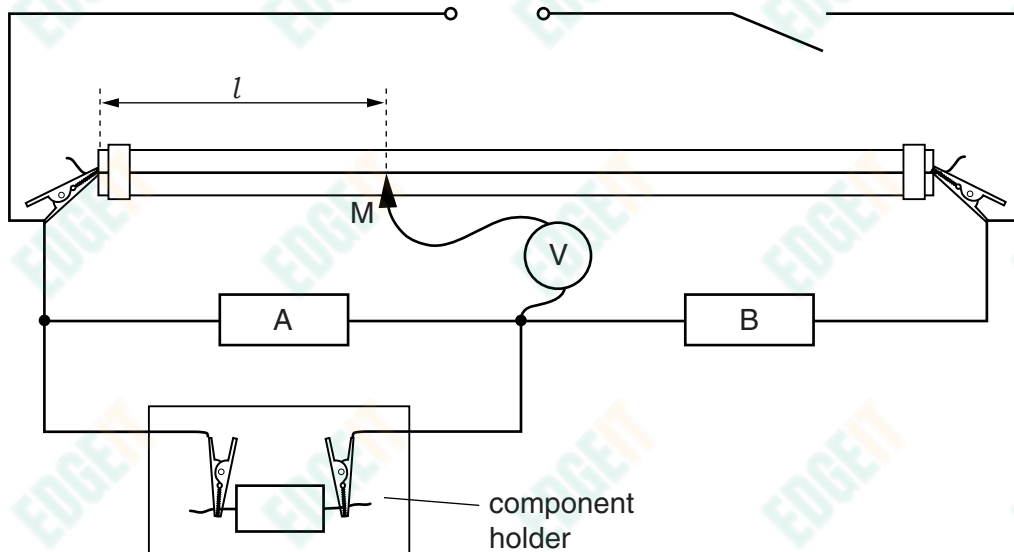


Fig. 1.2

- (ii) Close the switch. Adjust the position of M on the resistance wire until the voltmeter reads zero. Measure and record the length l as shown in Fig. 1.2.

$l = \dots\dots\dots [1]$

- (iii) Open the switch.

- (c) Repeat (b) using different labelled resistors in the component holder until you have six sets of values of R and l . Include values of $\frac{1}{R}$ and $\frac{1}{l}$ in your table.

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[10]

- (d) (i) Plot a graph of $\frac{1}{l}$ on the y -axis against $\frac{1}{R}$ on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

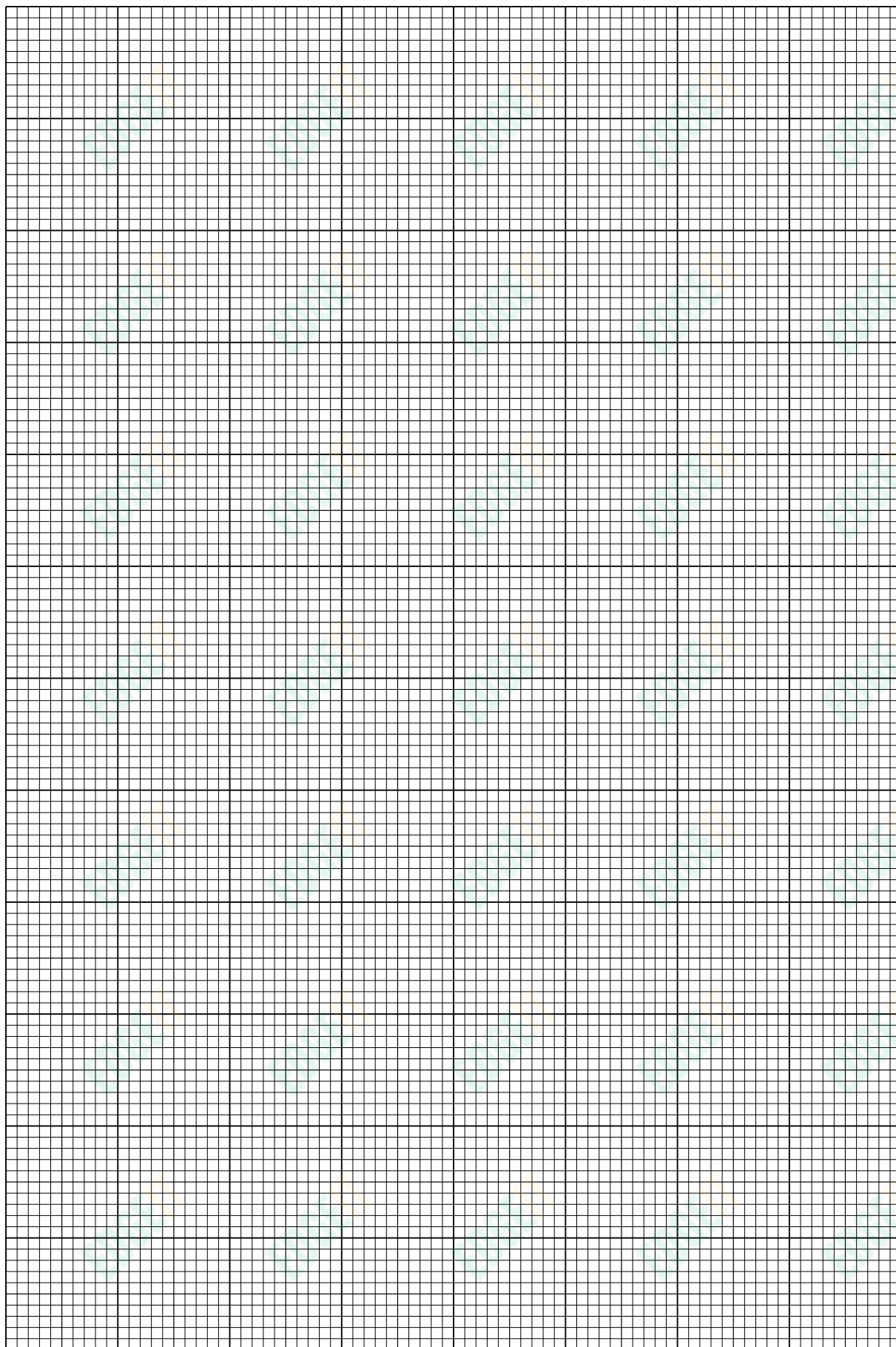
[1]

- (iii) Determine the gradient and y -intercept of this line.

gradient =

 y -intercept =

[2]



- (e) The quantities l and R are related by the equation

$$\frac{1}{l} = \frac{a}{R} + b$$

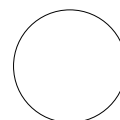
where a and b are constants.

Use your answers in (d)(iii) to determine the values of a and b .
Give appropriate units.

$a =$

$b =$

[2]



You may not need to use all of the materials provided.

2 In this experiment, you will investigate the motion of a partially-filled bottle.

- (a) (i)** Place the stand in the tray and suspend the empty bottle from the nail, as shown in Fig. 2.1.
Check that the bottle swings freely.

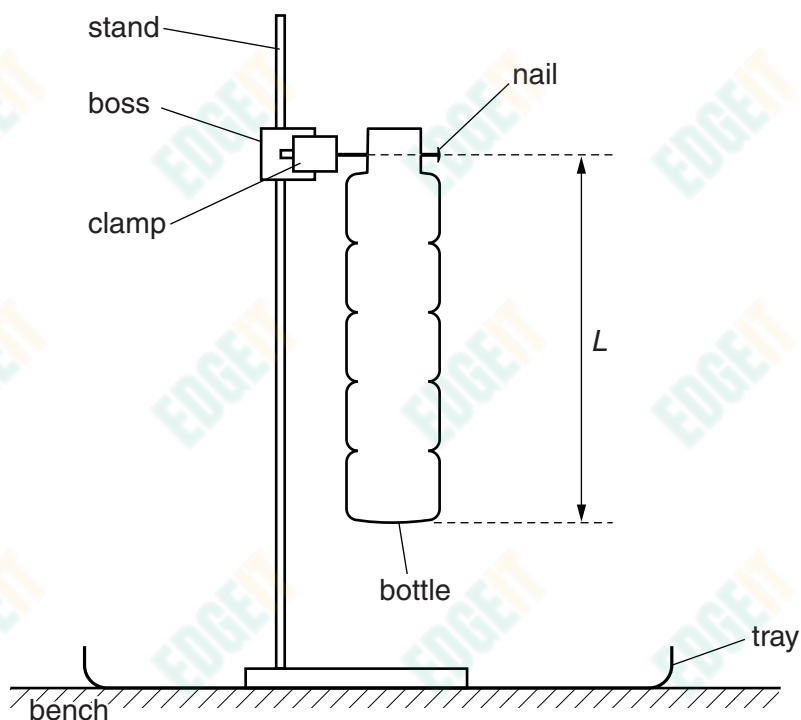


Fig. 2.1

- (ii)** Measure and record the distance L from the base of the bottle to the nail, as shown in Fig. 2.1.

$L = \dots\dots\dots$ mm [1]

- (b) (i) Pour water into the bottle until it is approximately half full, as shown in Fig. 2.2.

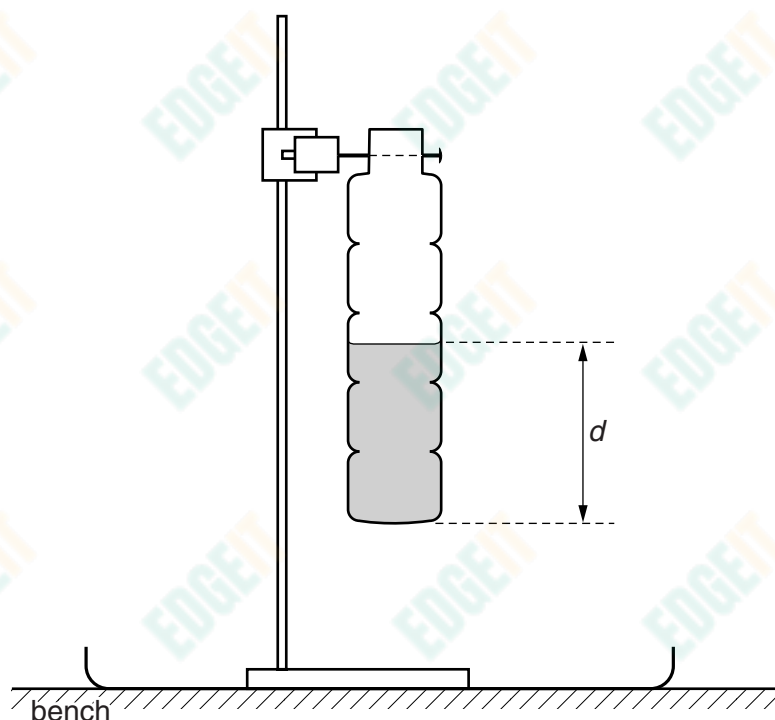


Fig. 2.2

- (ii) With the bottle hanging vertically, measure and record the distance d from the base of the bottle to the water surface, as shown in Fig. 2.2.

$d = \dots\dots\dots$ mm [1]

- (iii) Calculate the ratio c using the relationship $c = \frac{d}{L}$.

$c = \dots\dots\dots$

- (iv) Justify the number of significant figures that you have given for your value of c .

.....

 [1]

- (c) Calculate the value of q using the relationship

$$q = \frac{\frac{c^2}{3} - c + 1}{1 - \frac{c}{2}}.$$

$q = \dots\dots\dots$ [1]

- (d) (i) Push the base of the bottle approximately 4 cm to one side and then release it so that it swings, as shown in Fig. 2.3.

Ensure that the stand does not fall over when the bottle is swinging.

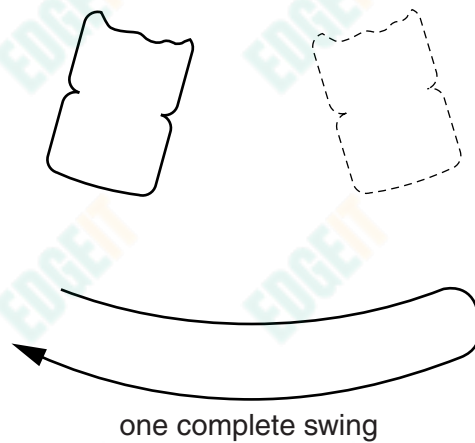


Fig. 2.3

- (ii) Measure and record the total time t for 10 complete swings.

$t = \dots\dots\dots$ [2]

- (iii) Estimate the percentage uncertainty in your value of t .

percentage uncertainty = $\dots\dots\dots$ [1]

- (e) (i) Pour more water into the bottle until it is approximately $\frac{3}{4}$ full.
 (ii) Repeat (b)(ii), (b)(iii), (c), (d)(i) and (d)(ii).

$d = \dots\dots\dots$ mm

$c = \dots\dots\dots$

$q = \dots\dots\dots$

$t = \dots\dots\dots$

[3]

- (f) It is suggested that the relationship between t and q is

$$t^2 = kq$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k = \dots\dots\dots$

second value of $k = \dots\dots\dots$

[1]

- (ii) Explain whether your results support the suggested relationship.

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..... [1]

- (g) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

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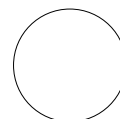
[4]



- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

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[4]



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